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To: Editor-in-Chief and Editorial Board

Journal: *IEEE Geoscience and Remote Sensing Letters (GRSL)*

Subject: Submission of Original Letter: *Hybrid Attention Transformers for Multi-Spectral Satellite Super-Resolution: Bridging Spatial Resolution and Radiometric Fidelity*

Dear Editor-in-Chief and Members of the Editorial Board,

Please find enclosed our original manuscript entitled “**Hybrid Attention Transformers for Multi-Spectral Satellite Super-Resolution: Bridging Spatial Resolution and Radiometric Fidelity**” for consideration for publication as a Letter in *IEEE Geoscience and Remote Sensing Letters (GRSL)*.

While spaceborne optical constellations like Copernicus Sentinel-2 provide vital multi-spectral coverage, aperture diffraction limits native resolution to 10 m Ground Sampling Distance (GSD). Applying conventional single-image super-resolution models designed for 8-bit photography to satellite imagery causes severe radiometric distortion, degrading biophysical metrics (such as NDVI) and blurring critical linear infrastructure.

In this letter, we propose **HAT-Light**, an edge-efficient continuous-scale Hybrid Attention Transformer engineered specifically for 4-channel (RGB+NIR), 16-bit Bottom-Of-Atmosphere (BOA) surface reflectance:

- **Radiometric Hybrid Attention Backbone:** Couples Shifted-Window Self-Attention with Depthwise Convolutional FFNs (DW-FFN), capturing global context while restoring localized translation-equivariant priors for runway and parcel geometries.
- **Continuous Scale Space Manifold:** Employs harmonic sinusoidal FiLM modulation to enable arbitrary continuous scaling ($s \in [2.0, 4.0]$) within a single unified checkpoint.
- **Composite Radiometric Physical Loss Suite:** Integrates Smooth Charbonnier regression, 2D rFFT spectral conservation, Sobel spatial gradients, and a numerically bounded Convex Cosine SAM loss that prevents gradient divergence in FP16.
- **Empirical Rigor & Real-Time Throughput:** Demonstrates state-of-the-art accuracy across 600 curated Sentinel-2 test patches (33.48 dB PSNR, 0.9048 SSIM, 1.37° SAM), biophysical NDVI conservation ($R^2 = 0.898$), and zero-shot transfer on 2.5 m USGS NAIP aerial imagery under the Wald protocol at **78.7 FPS** on edge GPUs.

This manuscript describes original, unpublished work not under concurrent review elsewhere. In compliance with IEEE preprint policy (PSPB Operations Manual Section 8.1.9), an earlier version has been deposited on engrXiv (Submission ID: 8320). All source code, pre-trained weights, and evaluation suites are publicly available at: <https://github.com/namans0071-code/HAT-Light-Satellite-Super-Resolution>.

Thank you for your consideration of our work.

Sincerely,

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